

Real time traffic measurements for high-speed networks

P C Hershey, C B Silio Jr and J G Wacławsky

BT requires timely information about high-speed network performance in order to manage these networks and to provide the best possible service to customers. Network management uses this information to evaluate, in real time, the state-of-network resources. Despite some claims to the contrary, traditional data collection techniques fail to support useful or accurate real time analysis because they either overflow available disk storage or exceed the real time processing capabilities of embedded processors. As network media speeds increase and as end users communicate across network architectures with differing protocols, the inadequacies of traditional techniques become more pronounced.

This paper presents a new information collection architecture (ICA) for real time traffic measurements that solves the data collection and analysis problem for high-speed networks. The ICA is applicable across network architectures and is scalable to SONET/SDH speeds and beyond. Also described is a prototype implementation of the ICA that provides performance measurements for token ring networks. This ICA prototype device includes programmable hardware to detect and collect traffic patterns of interest in real time, and an expert system to compute network performance using the collected data. Performance results computed using ICA prototype device traffic measurements taken from several operational token ring networks are presented and these results are compared with those computed using traffic measurements from a traditional network traffic monitor and with simulation results. These comparisons demonstrate that the ICA device accomplishes accurate performance measurement for real time network management by collecting only a small fraction of the data required by traditional approaches.

1. Introduction

Timely information about computer communications network performance is required in order to manage the network and to provide the best possible service to end users. This problem has been viewed as that of 'real time' network monitoring and performance evaluation [1]. Network monitoring includes the extraction, processing, collection, and presentation of dynamic information about the operation of a system [2]. Network management uses this information to evaluate in real time the state of network resources. Despite some claims to the contrary, traditional data collection techniques either overflow available disk storage or exceed the real time processing capabilities of embedded processors. These traditional techniques fail to support useful or accurate real time analysis for network management [3—7]. As network media speeds increase and as end users communicate across network architectures with differing protocols, the inadequacies of traditional techniques become more pronounced. High-speed media is defined as those that operate at greater than or equal to 10 Mbit/s.

Real time traffic measurements for performance evaluation of token ring networks are considered here. These measurements are used to compute utilisation of several operational token ring networks. A new network

management tool, called the information collection architecture (ICA) device, is presented that provides real time traffic measurement for token ring networks. The ICA device is used to measure performance of operating token ring networks, and these measurements are compared with measurements taken by a traditional network traffic monitor. Measured data from both approaches are also compared with simulation results. These comparisons show that the ICA device accomplishes accurate performance measurement for real time management by collecting only a small fraction of the data required by conventional approaches.

Traditionally, two approaches for traffic measurement are used:

- tracing and recording the actual data,
- collecting statistical information only.

The term 'trace' refers to a record of all frames and bytes transmitted on a network. Statistical information is parametric information that is usable in mathematical models for performance evaluation. The advantage of the statistical technique over trace techniques is that it stores a

small fraction of the data (e.g. only counter values). However, traditional statistical approaches have been inflexible, providing data for only one particular function or service usage.

The traffic measurement approach implemented with the ICA device is similar to traditional trace approaches for data collection in that the ICA device observes every bit of data that occurs on the communication media during the measurement time interval. Unlike trace approaches, the ICA device screens this data, and records only those portions required for performance evaluation. This screened data is called information. By collecting only this information, the disk storage limitation of trace techniques is not a problem for the ICA device. The ICA approach is similar to the traditional statistical approach in that the ICA device keeps counters for each parameter required for the performance measure of interest. However, unlike traditional statistical approaches, the ICA device can collect information for different network protocols, and for different network functions and services, by permitting the network service or function to specify the network protocol, the measurement time interval, and the required information to be collected from the network for performance evaluation. The ICA device accomplishes this using a unique combination of hardware and software to provide a data collection solution with the flexibility of software and the speed of hardware [8]. Software flexibility allows the ICA to collect data for arbitrary communication protocols and for varying measurement time intervals. Hardware speed provides the ability to capture data for high-speed network media. Such flexibility makes the ICA device as effective as trace techniques for viewing the data of interest and more adaptable than traditional statistical techniques.

Section 2 briefly describes the ICA device's hardware implementation. This implementation is used to demonstrate the feasibility, practicality, and utility of the ICA device for real time traffic measurement on IEEE 802.5 16 Mbit/s (and 4 Mbit/s) token ring networks. Section 3 presents definitions and mathematical models to express the performance measure utilisation in a form suitable for evaluation using traffic measurements obtained from the ICA device. These definitions and models are applied to the IEEE 802.5 token ring protocol. Section 4 presents and compares measured and simulation-derived utilisation results for several 16 Mbit/s token ring networks. Measured utilisation results are computed from data collection by both the ICA device and a traditional performance-measurement tool. Section 5 presents conclusions.

2. ICA hardware implementation

The ICA device implementation combines an expert system with a hardware card in order to measure performance of 16-Mbit/s or 4-Mbit/s token ring LANs [8]. Figure 1 shows a photograph of the hardware card called the pro-

grammable vector generator token ring (PVGTR) card. As shown in Fig 2, the PVGTR card consists of three main parts, a network interface (NI), an event-driven interface (EDI), and a micro-channel interface (MI).

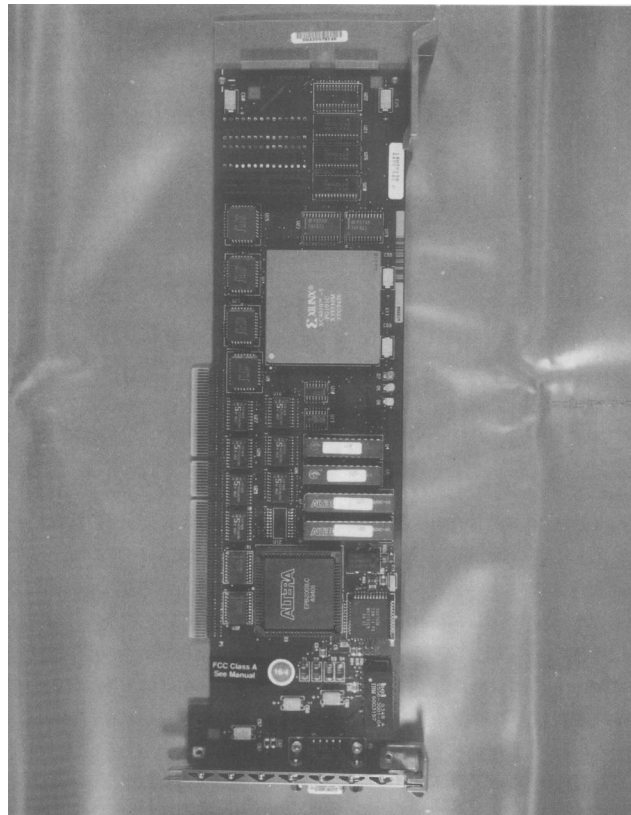


Fig 1 PVGTR card.

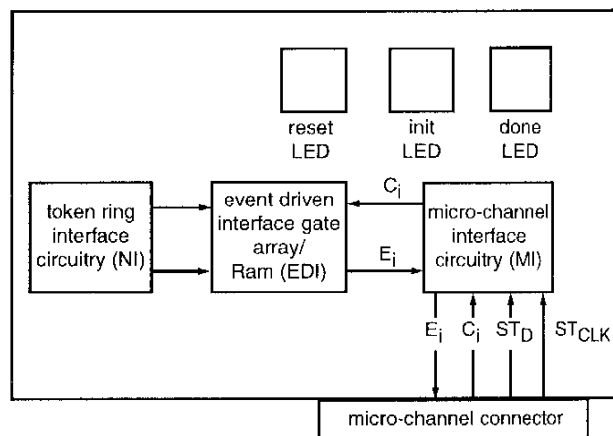


Fig 2 PVGTR card block diagram.

The NI provides the PVGTR card interface to the network by passively attaching to the token ring network. The NI is a real time monitor that inspects all activity of the communication process as it occurs, regardless of speed. The NI ensures that no 'required information' for performance assessment is lost.

The event-driven interface (EDI) is a programmable filter that screens the real time data for the minimum information required by the performance assessment process. The type of information collected by the EDI can be changed 'on-line' through the MI. The EDI includes pattern recognition finite state machines (FSMs) and pattern counters. The EDI was implemented using a XILINX 4005 field programmable gate array (FPGA) [9] for the pattern counters and using 64K bytes of static random access memory (SRAM) for the pattern recognition FSMs. State diagrams for the pattern recognition FSMs are presented elsewhere [8].

The micro-channel interface (MI) provides the programming interface for the EDI, and the MI sends the properly formatted information to the performance assessment process. The time frame in which to report the desired information to the performance assessment process depends on how this process uses the information. All control, status, and data transfer to and from the PVGTR card are done through the MI. A micro-channel interface was used in the initial ICA prototype device because the personal computers available for test purposes used a micro-channel bus. The MI and the micro-channel connector can be replaced by any other desired bus interface.

All PVGTR card components reside on a single IBM PS2 adapter card. The PVGTR card plugs into a 32-bit micro-channel connector of an IBM PS/2 model 70 or 80 personal computer. The PVGTR card communicates with the PS2 workstation over the micro-channel connector. A file of vectors, denoted C_i , passes from the workstation over the micro-channel connector, through the MI, and into the EDI. C_i contains the configuration data for the EDI. At the end of a measurement time interval the EDI outputs a vector of performance parameters that is denoted E_i . E_i passes through the MI, over the micro-channel connector, and to the workstation where it is used by the expert system to compute and assess performance.

The PVGTR card provides a functional self-test capability that tests all logic and interfaces except the actual token ring connection. As shown in Fig 2, self-test clock (ST_{CLK}) and self-test data (ST_D) are passed to the PVGTR card over the micro-channel connector. The PVGTR card also includes three light emitting diodes (LEDs) — the reset LED lights during PVGTR card reset, the init LED lights only if an EDI configuration error occurs, and the done LED lights after power-on until the EDI is fully configured.

3. Performance measures (utilisation)

The more frequently used performance measures include utilisation, throughput and delay [10]. Let BW denote bandwidth, which is the maximum number of bits that can be passed over a communications channel during a given time

period at the effective data rate. For example, on a 16-Mbit/s token ring network $BW = 16 \times 10^6$ bit/s. A bit time is the time required to transmit one bit at the effective bandwidth where one bit time is $1/BW$. For example, on a 16-Mbit/s token ring network, one bit time is $1/(16 \times 10^6 \text{ bit/s}) = 6.25 \times 10^{-8} \text{ sec/bit}$. Delay can be expressed in bits and can be converted to seconds by multiplying by the time per bit (i.e. the bit time) corresponding to a particular data rate.

This section investigates utilisation measures applied to the IEEE 802.5 token ring protocol in order to establish concrete examples for ICA device application. Analytical models that have appeared in the literature are considered for computing network utilisation, as well as some refinements to these models. Utilisation is the fraction of time spent in transmitting information bits out of the total time spent transmitting both information and overhead bits [10].

Let MTI denote the measurement time interval over which data is collected (in bits). The MTI can be converted to seconds by multiplying by $1/BW$ when the data rate is known, where BW is given in bit/s. In terms of an IEEE 802.5 token ring network, let TR denote a single token rotation on an idle ring; let T denote the number of token rotations during the MTI (called token count), let RL denote ring latency (in bits/ TR). Let $ATRT$ denote average token rotation time, which is the average number of bits required for a token to travel around a non-idle ring (i.e. a ring with user data on it) during the MTI . Thus, $ATRT = MTI/T$. Ring latency is the number of bits for a signal element (i.e. a channel symbol) to travel from its originating source station, around the entire ring, and back to its originating source station. Utilisation measures are then expressed in terms of these parameters.

Ring latency includes signal propagation delay (bits) through the ring medium plus the sum of the propagation delay (bits) through each station connected to the ring, plus transmission/reception synchronisation delay (bits) [11]. In an IEEE 802.5 token ring network, one station serves as the active monitor that both provides the master clock signal and maintains synchronisation for the ring. It is possible for segments of the network to operate at speeds slightly higher or lower than the master clock. The combined effect of these variations for all stations on the ring can cause variations in ring latency. The active monitor places an elastic buffer in the data path to absorb these variations. The size of this elastic buffer for a 16 Mbit/s token ring network is initialised to 40 bits [11]. Each bit in the elastic buffer causes a 1-bit delay on the token ring network. If the received signal at the active monitor is slightly faster than the master clock, then the elastic buffer expands as needed to maintain constant ring latency. Likewise, if the received signal is slower than the master clock, then the buffer contracts to maintain constant ring latency. The length of the elastic buffer is added to the fixed 24-bit buffer that

corresponds to the number of bits in a token. The propagation delay through each active station also increases ring latency. For the 16 Mbit/s token ring network each active station causes a 2.2-bit propagation delay [12]. A final element of ring latency is the signal propagation delay through the ring medium. Let P_{md} denote this delay (bits), and let L_T denote the total network cable length in metres (m) traversed during a single token rotation (TR). Assume the use of shielded twisted pair cable with signal propagation velocity of 2×10^8 m/sec, which delays each bit by 5×10^{-9} sec/m [12]. Then P_{md} is given by:

$$P_{md} = L_T \times 5 \times 10^{-9} \times BW \text{ bits/TR} \quad \text{..... (1)}$$

For a 16-Mbit/s token ring network, $P_{md} = L_T \times 5 \times 10^{-9} \times 16 \times 10^6 \text{ bits/TR} = L_T \times 8 \times 10^{-2} \text{ bits/TR}$. The total ring latency for a token ring network with N nodes is

$$RL = (24 + 40 + 2.2 \times N + L_T \times 5 \times 10^{-9} \times BW) \text{ bits/TR} \quad \text{.....(2)}$$

Total ring latency is expressed in seconds for a 16-Mbit/s token ring network with N nodes by multiplying by a bit time, that is:

$$RL = (64 + 2.2 \times N + L_T \times 8 \times 10^{-2}) \times 6.25 \times 10^{-8} \text{ sec/TR} \quad \text{..... (3)}$$

Alternatively, RL can be measured directly on the ring by measuring the number of bits of delay between the end of one token and the end of a consecutive token, provided there are no other intervening frames.

Three different utilisation measures from the literature are now examined for computing utilisation on token ring networks [12—14]. These three measures are labelled U_{T1} , U_{T2} and U_{T3} . These measures are applicable to any token ring network regardless of the media speed. In the presentation below, a media speed of 16 Mbit/s is considered.

U_{T1} was originally proposed by Ulm [13] and later used by others [15,16]. U_{T1} is stated in terms of per cent utilisation with ring latency (RL in bits per token rotation from equation (2)) and average token rotation time ($ATRT$ in bits per token rotation). The expression for U_{T1} is:

$$U_{T1} = 100(1 - RL/ATRT)\% = 100(1 - RL \times T/MTI)\% \quad \text{..... (4)}$$

If the number of stations N and the cable length L_T remain constant during the measurement time interval, then equation (2) can be used to compute RL . MTI is a parameter chosen by the analyst. Therefore, the only 'information' that must be collected from the network is the token rotation count T in order to compute $ATRT$.

U_{T2} was originally proposed by Irvin [12]. U_{T2} is stated in terms of per cent utilisation. Let F denote the frame count in an MTI (with MTI expressed in bits). Let Mst_{T2} denote the mean service time per frame (in bits/frame). The expression for U_{T2} is:

$$U_{T2} = 100(F \times Mst_{T2})\% \quad \text{..... (5)}$$

Let IFS_{T2} (in bits/frame) denote the mean size of the frame information field for the MTI ; let $NIFS_{T2}$ (in bits/frame) denote the mean size of all other frame fields (i.e. frame overhead) for the MTI . Then Mst_{T2} is given by the expression:

$$Mst_{T2} = (NIFS_{T2} + IFS_{T2}) \text{ bits/frame} \quad \text{..... (6)}$$

Although the IEEE 802.5 token ring standard frame format indicates that the number of overhead (i.e. non-information) bytes per frame is 22 [11], Irvin [12] suggests that the use of 21 bytes instead of 22 bytes per frame in the expression for $NIFS_{T2}$ yields a more accurate measure of utilisation using U_{T2} . Thus, at 8 bits/byte $NIFS_{T2} = 8 \times 21 \times 6.25 \times 10^{-8} \text{ sec/frame}$ on a 16-Mbit/s token ring.

A new expression is now presented to estimate IFS_{T2} for the IEEE 802.5 token ring network. Because the IEEE 802.5 token ring networks investigated do not use early token release [11], an inter-frame gap (IFG) must be accounted for. The IFG is defined as the number of bits between the observation of the sixth bit of the ending delimiter (ED) field of the present frame or token circulating around the ring and the observation of the first bit of the starting delimiter (SD) field of the next frame or token to circulate around the ring. An IFG exists between consecutive tokens, consecutive frames, frames followed by tokens, and tokens followed by frames. The IFG depends on the number of active stations N , the size of the active monitor elastic buffer, and the signal propagation delay through the ring medium and through each active station. Let IFG_T denote the inter-frame gap between consecutive tokens, and let IFG_F denote the inter-frame gap between consecutive frames, or between frames followed by tokens, or between tokens followed by frames. Assuming that BW is 16 Mbit/s, that there are N active stations during the MTI , that the signal propagation delay through the ring medium equals $L_T \times 8 \times 10^{-2} \text{ bits per token (or per frame) rotation}$, that the propagation delay equals 2.2 bits per token (or per frame) per active station [12], and that the delay caused by the elastic buffer initially equals 40 bits per token (or per frame) [11], then IFG for a 16-Mbit/s token ring is given by:

$$IFG = IFG_T + IFG_F, \text{ where}$$

$$IFG_T = (40 + 2.2 \times N + L_T \times 8 \times 10^{-2}) \text{ sec/token rotation, and}$$

$$IFG_F = (40 + 2.2 \times N + L_T \times 8 \times 10^{-2}) \text{ sec/frame rotation.} \quad \dots (7)$$

The IFG value of equation (7) varies with the number of stations, with changes in the elastic buffer size, with changes in the length of the ring medium, and with changes in the propagation delay per active station. For a token ring network with a constant number of stations, constant propagation delay, and constant elastic buffer size, equation (7) provides a constant IFG value between consecutive tokens, between consecutive frames, between tokens and frames, and between frames and tokens.

To solve for IFS_{T2} the total number of bits for the MTI is reduced by the number of bits required for ring latency (i.e. $T \times RL$), by the number of bits for the inter-frame gap IFG_F (i.e. $F \times IFG_F$), and by the number of bits for the frame overhead (i.e. $F \times NIFS_{T2}$). (Note that RL includes the inter-frame gap for consecutive tokens, IFG_T .) This result is divided by the frame count (i.e. F). Thus, IFS_{T2} is:

$$IFS_{T2} = [MTI(1 - (T \times RL)/MTI - (F \times IFG_F)/MTI - (F \times NIFS_{T2})/MTI)]/F \text{ bits/frame,} \quad \dots (8)$$

which on a 16-Mbit/s token ring network can be converted to sec/frame by multiplying by a bit time. To compute U_{T2} in equation (5), T , MTI , RL , IFG_F , and F must be known. MTI is a chosen parameter, RL and IFG_F are calculated (assuming a constant number of stations on the ring and constant propagation delay during the MTI); and the only parameters that must be measured from the ring are the counts T and F .

The utilisation models presented in equations (4) and (5) have several limitations. As pointed out by Ulm [13], utilisation approximation U_{T1} of equation (4) is not appropriate for computing utilisation in all applications. To investigate the limitations of measure U_{T1} , a simulation program has been written with deterministic arrival and frame size input parameters [8]. For one simulation test case, utilisation was computed for logical link control (LLC) [11] frame data only during an MTI . The following parameter values were used: $BW = 16 \times 10^6$ bits/sec, $MTI = 2925/BW = 1.828 \times 10^{-4}$ sec, $RL = 146.237$ bits/TR (i.e. with $P_{md} = 7.437$ bits, with $N = 34$ active stations, with constant propagation delay of 2.2 bits per station, and with constant elastic buffer delay of 40 bits), $T = 16$ TR (i.e. 16 tokens were observed during the MTI), and $F = 0$ (i.e. an idle ring). Substituting these parameter values in equation (4) yields:

$$U_{T1} = 1 - ((16 \times 146.237 \times 6.25 \times 10^{-8})/(1.828 \times 10^{-4})) = 20\% \quad \dots (9)$$

However, with a frame count of $F = 0$, the ring is idle with respect to LLC frame data, and the LLC utilisation should equal 0%; thus U_{T1} provides an erroneous prediction in this case. This error occurs because non-LLC ring

activity, such as tokens, MAC frames, and other environmental data are not accounted for in U_{T1} . Figure 3 presents additional U_{T1} results obtained from the simulation with deterministic arrivals, by varying the number of frames transmitted and by keeping all other parameter values unchanged from above. In order to overcome the erroneous predictions of utilisation, a utilisation measure that includes frame count (F) as a parameter is required.

The U_{T2} prediction in equation (5) includes the frame count as a performance parameter. Simulation results presented elsewhere [8] verify that U_{T2} provides the correct LLC frame data utilisation results for the idle ring. However, U_{T2} does not correctly predict utilisation for MAC frame data only, nor for non-MAC frame data only (i.e. LLC frame data plus environmental data).

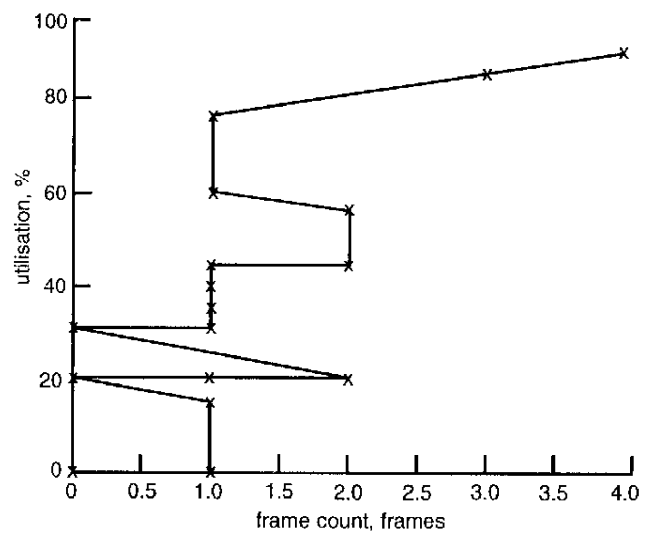


Fig 3 Average utilisation (%) versus frame count. (Token ring simulation — curve corresponds to utilisation measure U_{T1} .)

U_{T3} represents the traditional method [14] for measuring utilisation. Let FB denote the total number frame information bits transmitted during the MTI . Frame information bits are the bits contained within the information field of the MAC frame format [11].

$$U_{T3} = 100 (FB/MTI)\% \quad \dots (10)$$

In traditional trace techniques [14] all bits appearing on the ring are collected during the MTI , including all of the frame information bits.

4. Test and validation results

This section presents results for 16-Mbit/s token ring network utilisation that were computed using both simulated and measured network traffic information. Even though the ICA device can collect information on both 4-Mbit/s and 16-Mbit/s token ring networks, it has been

used to collect information on 16-Mbit/s networks only because this higher speed network better challenges the capabilities of the ICA device. The ICA device has been used to collect data from several different operational 16-Mbit/s token ring networks. Data has been collected for deterministic traffic generated on 4-node token ring networks located in Manassas, Virginia, USA, and in Gaithersburg, Maryland, USA. Actual traffic data has also been collected from a 19-node token ring network in a production environment located in Gaithersburg, Maryland. This data has been used to compute utilisation according to performance measures that are suitable for the ICA method. On this same data, a conventional LAN measurement tool [14] has also been used to determine utilisation.

For all results sample mean utilisation is plotted with 95% confidence intervals [17]. Henceforth the sample mean of utilisation is referred to as 'average utilisation'. Recall that U_{T1} in equation (4) [13] provides a measure of utilisation based on collecting only the token rotation count per *MTI*, U_{T2} in equation (5) [12] provides a measure of utilisation based on both the token rotation count and the frame count per *MTI*, and U_{T3} in equation (10) [14] provides a measure of utilisation based on counting the number of frame information bits per *MTI*.

In order to minimise ICA device memory and counter sizes for collecting required performance parameters while at the same time maintaining measurement accuracy, U_{T2} pattern detection was chosen for implementation because U_{T2} provides better measurement accuracy than does U_{T1} . Therefore U_{T2} results are presented in this section.

4.1 Simulation results

Two simulations have been written for traffic on an IEEE 802.5 token ring network — one for random times between transmitted frames and random frame sizes, and one for deterministic times between transmitted frames and deterministic frame sizes [8]. In both simulations 16-Mbit/s token ring networks are assumed, and $MTI = 16 \times 10^6$ bits (i.e. 1 second) is chosen. Although the ICA method and these simulations apply to any speed network, the 16-Mbit/s token ring network has been simulated in order to compare simulated results with those computed using ICA device measurements.

The simulations collect data as the ICA device would collect data, and compute utilisation according to performance measure U_{T2} in equation (5). They also collect data as the trace device [14] would collect data, and compute utilisation according to performance measure U_{T3} in equation (10).

4.1.1 Simulation with random arrivals and random frame sizes

It is assumed that frames arrive at each node for transmission on the network according to independent Poisson processes, and also that the size of frames in the Poisson arrival streams at each node are determined by independent, exponentially and identically distributed random variables with common mean $E[FB]$. The exponentially distributed frame sizes are truncated to integer values, resulting in geometrically distributed frame sizes [18]. The sample mean of utilisation is computed with 95% confidence intervals [17]. In Fig 4, these results are plotted for average utilisation (%) versus average number of frames transmitted per second. Error bars are used in the plots to illustrate the 95% confidence intervals. Each data point represents the sample mean of 30 samples. Average frame sizes were held approximately constant, varying by less than 0.5%, for each point on the plot. For utilisation below 50%, the simulation results for random arrivals, computed using U_{T2} in equation (5) (curve A), agree within the 95% confidence interval with those computed using U_{T3} in equation (10) (curve B). Thus, there is no statistically significant difference between the results computed using U_{T2} and those computed using U_{T3} . For utilisation above 50%, the 95% confidence intervals about the sample mean for the ICA results computed using U_{T2} in equation (5) do not overlap those computed using U_{T3} in equation (10). One reason for this is that Irvin used 21 bytes for the number of overhead bytes per frame in equation (5) rather than 22 overhead bytes per frame as called for by the IEEE 802.5 standard. This small reduction in frame size is not distinguishable when utilisation is below 50% for the random arrivals. For utilisation above 50%, Irvin's approximation of fewer overhead bytes results in a higher calculated arrival rate and a correspondingly higher

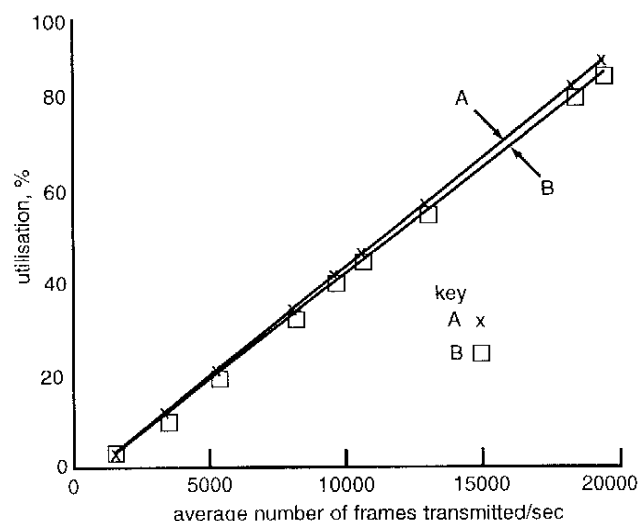


Fig 4 Average utilisation (%) versus average number of frames transmitted per second. (Token ring simulation with random arrivals — curve A corresponds to utilisation measure U_{T2} , curve B corresponds to utilisation measure U_{T3} .)

calculated utilisation than would the use of the IEEE 802.5 standard overhead specification of 22 bytes.

4.1.2 Simulation with deterministic arrivals and deterministic frame sizes

At each node a known number of frames per second and known frame sizes are generated to produce utilisations of 5%, 29%, 55% and 97%. A 4-node, 16-Mbit/s token ring network is assumed with each node attached to the ring via 2.734 m shielded twisted pair cable. Using equation (3) a ring latency is computed for this simulated token ring network with deterministic arrivals and frame sizes, denoted RL_{DS} , as:

$$RL_{DS} = (24 + 40 + 2.2 \times 4 + 2.734 \times 8 \times 10^{-2} \times 4) \times 6.25 \times 10^{-8} \text{ sec/TR} \\ = 4.61 \times 10^{-6} \text{ sec/TR} \dots (11)$$

The maximum *MTI* for simulated data collection was 1.5 hours. This *MTI* corresponds to the maximum *MTI* over which the 30-bit token counter of the ICA device can collect token counts on an idle ring.

The sample mean of utilisation is computed with 95% confidence intervals [17]. Figure 5 plots these results for average utilisation (%) versus frame size. Figure 6 plots average utilisation (%) versus number of frames transmitted per second. In all cases, the ICA results computed using U_{T2} in equation (5) (curve A) match those computed using equation (10) (curve B) within the 95% confidence interval about the mean of the ICA results. In Fig 5 it can be observed that utilisation increases with increasing frame size. This is expected because larger frame sizes mean a larger information field and relatively fewer overhead bits per second than do smaller frames, and overhead bits reduce U_{T2} utilisation. Figure 6 shows that utilisation decreases with an increasing number of frames per second. These results differ from those in Fig 4 where the frame sizes were assumed to be fixed. Here, higher utilisation results from transmitting fewer frames with large information fields and relatively few overhead bits, and lower utilisation results from transmitting more frames with smaller information fields and a relatively large number of overhead bits.

4.2 Measured results

This section presents the measured results obtained using the ICA device. Data was collected on two 4-node, 16-Mbit/s token ring networks on which deterministic traffic was transmitted, and on a 19-node, 16-Mbit/s token ring network with actual traffic patterns in an industrial environment. Each node was attached to its respective ring via a 2.734 m shielded twisted pair cable. Utilisation is computed for these three networks using U_{T2} for the ICA device and using U_{T3} for the

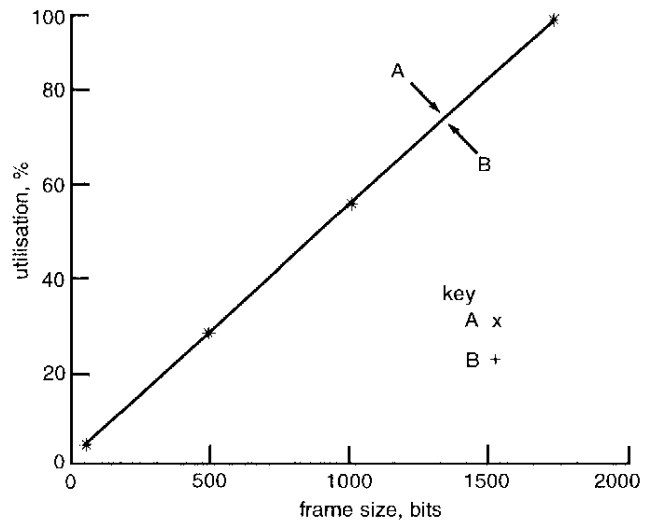


Fig 5 Average utilisation (%) versus frame size. (Token ring simulation with deterministic arrivals — curve A corresponds to utilisation measure U_{T2} , curve B corresponds to utilisation measure U_{T3} .)

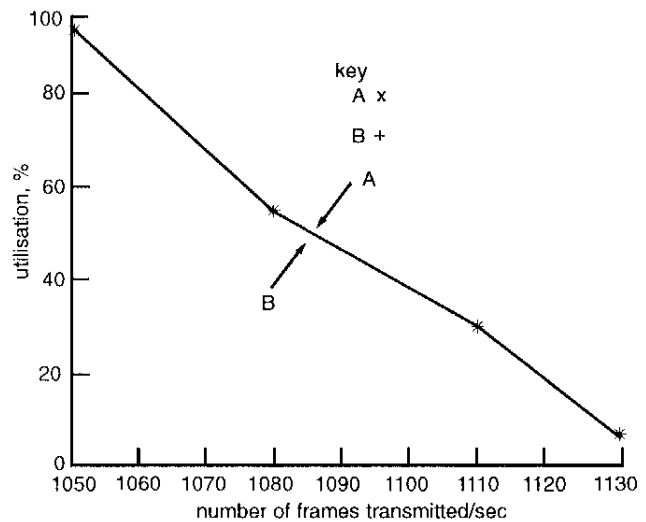


Fig 6 Average utilisation (%) versus number of frames transmitted per second. (Token ring simulation with deterministic arrivals — curve A corresponds to utilisation measure U_{T2} , curve B corresponds to utilisation measure U_{T3} .)

traditional LAN analysis trace device [14]. The ICA device and trace device measured results are compared for utilisation, as are the measured results with simulation results.

4.2.1 Four-node tests, Manassas, Virginia

The Manassas 4-node token ring LAN configuration is shown in Fig 7. Each node is a station that interfaces to the ring via a multi-station access unit (MSAU). One station contains the ICA device, two stations contain traffic generators, and one station contains the trace device. The traffic generators are configured as 16-Mbit/s token ring adapters that send traffic on to the ring. The number of

frames transmitted per second and the frame size must be set to a given level in the traffic generators at the start of a test. Thus, both the number of frames transmitted per second and the frame size are deterministic. In these experiments, frame sizes (in units of bytes/frame) are generated according to the information in Table 1. In Table 1 TG1 refers to traffic generator 1, and TG2 refers to traffic generator 2.

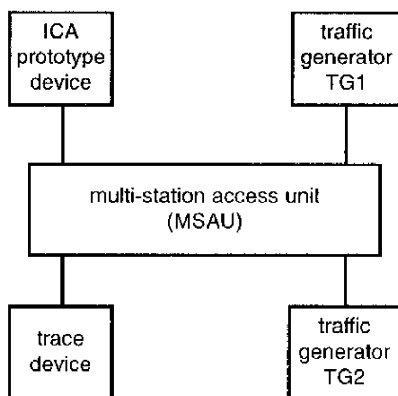


Fig 7 4-node LAN configuration.

Table 1 4-node traffic generator frame sizes (bytes/frame).

Utilisation (%)	TG1 (bytes/frame)	TG2 (bytes/frame)
97	2200	1650
93	2218	1512
71	1500	1250
65	1200	1200
55	1000	1000
40	750	750
34	600	600
29	500	500
22	450	350
15	250	250
5	50	75
2.2	10	10

In these experiments, the trace device is also used [14] to measure utilisation. The trace device measures ring latency in addition to utilisation. In order to do so, however, the trace device generates a 5389-byte frame every other second. These frame bytes add to the overhead of the channel. Thus, the trace device is not a passive monitor as is the ICA device.

In Fig 8 average utilisation (%) is plotted against MTI for a group of deterministic results measured using the Manassas laboratory set-up. Average utilisation refers to the sample mean of utilisation shown with 95% confidence intervals [17]. As in the simulation plots, each data point represents the sample mean of 30 samples. In Fig 8 it can be seen that average utilisation results obtained from the trace device, using measure U_{T3} of equation (10) (curves B, D, F and H), are lower than those obtained from the ICA device, using measure U_{T2} of equation (5) (curves A, C, E and G). Note that the pairs of curves (A, B), (C, D), (E, F), and (G, H) seem farther apart at low utilisation and get closer at high utilisation (greater than 90%). The separation in utilisation measurements for corresponding pairs of curves is caused by differences in the way the trace device and the ICA device work. The trace device itself transmits an extra 5389-byte frame every other second, and the ICA device is completely passive. The trace device ignores this extra frame for utilisation computation. The ICA device considers this extra frame to be part of the network traffic and includes it in the utilisation computation because it contributes to network load. For the Manassas 4-node tests higher utilisation levels correspond to the transmission of larger frames in the MTI, and the extra trace device frames have less effect on the U_{T2} computation. Likewise, lower utilisation levels correspond to smaller frame sizes transmitted in the MTI, and the extra trace device frames show up as a higher utilisation level in the U_{T2} computation.

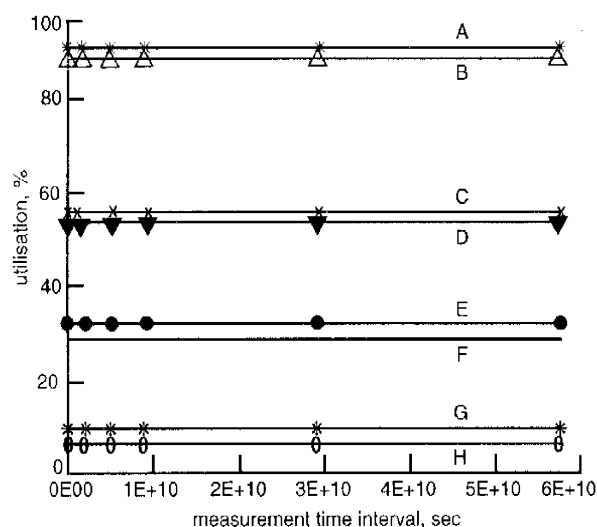


Fig 8 Average utilisation (%) versus measurement time interval. (Measured tests on 4-node, 16-Mbit/s token ring with deterministic arrivals and frame sizes — curves A, C, E, and G correspond to ICA device measured data using utilisation measure U_{T2} , curves B, D, F and H correspond to trace device measured data using utilisation measure U_{T3} .)

4.2.2 19-node and 4-node tests, Gaithersburg, Maryland

The 19-node, 16-Mbit/s token ring network on which the measurements were taken is used by engineers in Gaithersburg, Maryland. Utilisation results were computed

using the information collected by both the ICA device and the trace device.

In Fig 9, average utilisation (%) is plotted against the average number of frames transmitted per second, measured by the ICA device and computed using U_{T2} both with and without the trace device operating in the network. Each data point results from 30 samples corresponding to counts taken during 30 contiguous MTIs of one second each. Data was taken at various times of day over a week. Notice that when the trace device is operating in the network, the data is less scattered (data set A) than when the trace device is not operating (data set B). This results from the extra 5389-byte frame that the trace device puts on to the network every other second.

For both data sets, average utilisation increases with an increasing average number of frames transmitted per second. The trend shown in these results is similar to that shown (in Fig 4) by simulation results under random arrivals and random frame sizes.

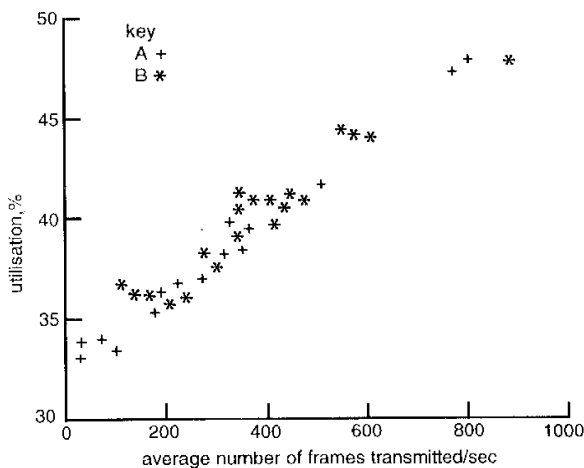


Fig 9 Average utilisation (%) versus average number of frames transmitted per second. (Measured tests on 19-node, 16-Mbit/s token ring with random arrivals. Data set A corresponds to ICA device measured data using utilisation measure U_{T2} with the trace device running during traffic measurement. Data sets B corresponds to ICA device measured data using utilisation measure U_{T2} with the trace device not running during traffic measurement.)

In a second experiment at Gaithersburg, deterministic traffic was transmitted over a 4-node token ring network. In Fig 10, measured results from these tests (curve B) are compared with those from the Manassas 4-node test (curve A). In Fig 10 the relationship between average throughput and average utilisation can be seen. It can also be seen that as throughput increases, so does utilisation, where throughput equals the product of the average frame size times the average frame departure rate from the transmitting traffic sources. Throughput here refers only to user data traffic appearing on the ring.

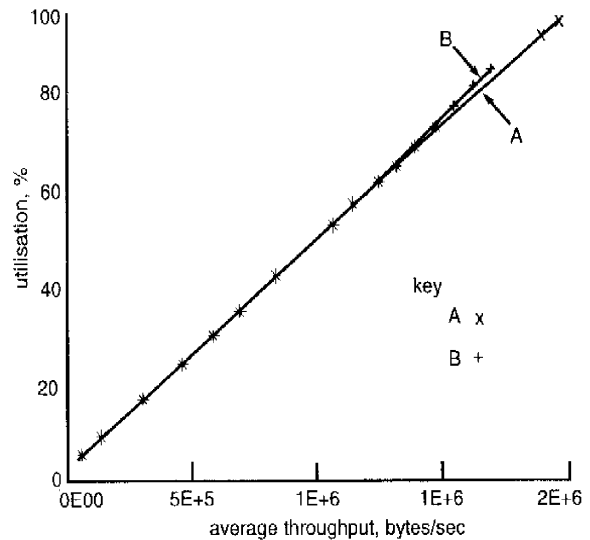


Fig 10 Average utilisation (%) versus average throughput. (Measured test on 4-node, 16-Mbit/s token rings with deterministic arrivals and frame sizes. Curve A corresponds to ICA device measured data using utilisation measure U_{T2} for the Manassas 4-node token ring. Curve B corresponds to ICA device measured data using utilisation measure U_{T2} for the Gaithersburg 4-node token ring.)

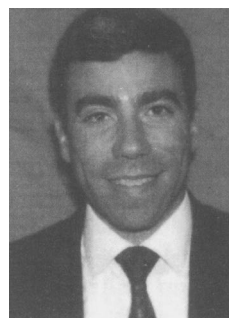
5. Conclusions

Real time traffic measurement techniques for performance evaluation of token ring networks have been investigated. Three performance measures for computing utilisation have been presented and compared — U_{T1} which is based on token count, U_{T2} which is based on both token and frame counts, and U_{T3} which is based on frame bit counts. Utilisation results have been calculated using these measures for data collected from token ring networks with both simulated random traffic and simulated deterministic traffic, from two separate operational token ring networks with deterministic traffic generators, and from a token ring network in a production environment with actual traffic. These results indicate that measure U_{T2} is suitable for real time traffic measurement devices such as the information collection architecture (ICA) device. The ICA device described here provides real time traffic measurement for both 16-Mbit/s and 4-Mbit/s token ring networks. ICA device results using U_{T2} have been compared with traditional trace-device results using U_{T3} and with simulation results. These comparisons have shown that the ICA device using U_{T2} accomplishes performance measurement for real time network management as accurately as do traditional approaches using U_{T3} . Because the ICA device collects only a small fraction of the data required by those approaches, it significantly reduces the danger of overflowing available storage or exceeding the real time processing capabilities of embedded processors. Furthermore, because the ICA device attaches passively to the network, it does not reduce the network capacity for transmitting user data. It is anticipated that the ICA device will soon become part of commercial products.

This paper has briefly described a prototype implementation of the ICA for performance measurement of token ring networks. Because the ICA device uses programmable hardware, it can be reprogrammed to collect performance measures for other networks. An extension of the ICA to collect performance parameters for high-speed wide area networks (e.g. SONET and SDH) is presented elsewhere [8].

References

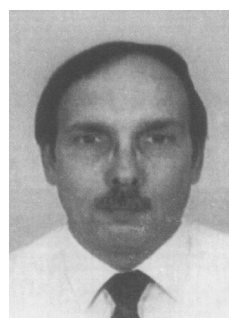
- 1 Mazumdar S and Lazar A A: 'Monitoring integrated networks for performance management', IEEE Proc ICC 90, Atlanta, GA, 2, pp 289—293 (April 1990).
- 2 Snodgrase R: 'A relational approach to monitoring complex systems', ACM Trans Comp Sys, 6, No 2, pp 157—196 (May 1988).
- 3 Tolly K: 'Token ring analysers — new tools for fighting network wildfires', Data Communications Magazine, 20, No 1, pp 17—37 (November 1993).
- 4 Tolly K, Curtin J C and Hindin E M: 'What's missing from your LAN analysers', Data Communications Magazine, 22, No 2, pp 41—55 (January 1993).
- 5 Koontz C P: 'LAN Times compares analysers', LAN Times Magazine, 10, No 2, pp 53—73 (January 1992).
- 6 Montgomery J and Maxwell K: 'Network analysers', Corporate Computing, 1, No 4, pp 113—133 (August 1992).
- 7 Coffee P: 'Expert sniffer makes diagnoses — and errors', PC Week Networking, pp 47—50 (April 1992).
- 8 Hershey P C: 'Information collection architecture for performance measurement of computer networks', PhD Dissertation, University of Maryland, College Park, MD (May 1994).
- 9 XILINX Inc: 'XILINX Programmable Gate Array Data Book', San Jose, CA (1991).
- 10 Abeyesundara B W and Kamal A E: 'High-speed local area networks and their performance: a survey', ACM Computing Surveys, 23, No 2, pp 221—264 (June 1991).
- 11 IEEE Inc: 'IEEE Standard 802.5, Token Ring Access Method', New York, NY (1989).
- 12 Irvin D R: 'An introduction to the transmission performance capabilities of IEEE 802.5 token ring networks', Computer Communication Review, ACM Press, New York, NY, pp 25—34 (October/November 1987).
- 13 Ulm J M: 'A timed token ring local area network and its performance characteristics', IEEE Proc 7th Conf Local Comput Networks, Minneapolis, MN, pp 50—56 (February 1982).
- 14 IBM Corp: 'IBM Token-Ring Network 16/4 Trace and Performance Program', Research Triangle Park, NC (June 1990).
- 15 Barker K J and Lingafelt C S: 'Programmable digital filter', IBM Technical Disclosure Bulletin, 31, No 2, pp 198—204 (July 1988).
- 16 Yang Y Y and Sankar R: 'Maximising FDDI performance by parameter tuning', IEEE Proc INFOCOM 91, Bal Harbour, FL, pp 1431—1439 (April 1991).
- 17 Kobayashi H: 'Modelling and analysis: an introduction to system performance evaluation methodology', Addison-Wesley, Reading, MA (1978).
- 18 Ghafir H M, Silio C B and Wasser E M: 'An improved model for service time on PLAYTHROUGH rings', IEEE Proc Computer Networking Symp, Washington DC, pp 87—95 (April 1988).



Paul C Hershey joined the US Systems Engineering Centre (USSEC) of BT in 1994 as a Senior Member of the Technical Staff where he is responsible for analysis of SONET, SDH, ATM, and broadband network management products, services, and architectures. He earned a BA in Mathematics from the College of William and Mary in 1980, and MS and PhD degrees both in Electrical Engineering from the University of Maryland in 1984 and 1994, respectively. He has authored patents and publications in the areas of communications network performance analysis, network management, security, routing, fault tolerance, and multimedia. From 1980 to 1994, he worked for IBM, Fairchild, and BDM, architecting, designing and developing communications products and systems. He is a member of IEEE.



Charles B Silio Jr earned BSEE, MSEE, and PhD degrees in electrical engineering at the University of Notre Dame in 1965, 67, and 70. He served in the US Army and in 1972 joined the Electrical Engineering Dept, University of Maryland, College Park, where he is an Associate Professor. During 1985—87 he was Associate Department Chair for Graduate Studies. His work is in computer engineering and digital systems design, and his research interests include design, performance evaluation, and reliability analysis of computer communications networks, computer security, and multiple-valued logic. He has been active in providing continuing education for engineers and has received awards for contributions to electrical engineering education. He is a member of Eta Kappa Nu, Tau Beta Pi, Sigma Xi, and is a senior member of IEEE.



John G Waclawsky earned a PhD from the University of Maryland. He does research and development for IBM services.

He is active in research and consulting on issues related to performance analysis, expert systems, fault tolerance, LAN and WAN protocols and high-speed network environments.